

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I DOMA SAKETH-192525018____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- a. Encrypt using ECB and CBC modes (Assume IV = 0000)
- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

1. RSA Encryption Analysis

Given:

-
-
-
- Message

(a) Compute and
Step 1: Compute

Step 2: Compute Euler's Totient

Answer:

-
-

(b) Find the private key

We need

Given

Find inverse of 7 modulo 120.

Hence

Public Key: (7,143)

Private Key: (103,143)

(c) Encrypt the message

Formula

Using modular arithmetic,

Ciphertext

(d) Decrypt

Formula

After modular exponentiation,

Original message is recovered.

Verification

Original Message = 9

Recovered Message = 9 ✓

(e) Effect of changing

- Small (3 or 5)
 - Faster encryption
 - May introduce vulnerabilities if padding is weak.
- Moderate
 - Industry standard
 - Fast and secure.
- Very large
 - Slower encryption
 - Slightly stronger against brute-force attacks but not commonly used.

Conclusion: provides the best balance of security and performance.

2. Block Cipher Mode Evaluation

Given

Plaintext blocks

Key

IV

Encryption operation:

(a) ECB Mode

Block 1

1010 XOR 1100
=0110

Block 2

1010 XOR 1100
=0110

Block 3

1111 XOR 1100
=0011

Block 4

1010 XOR 1100
=0110

ECB Ciphertext

CBC Mode

Formula

Initial

Block 1

1010 XOR0000

=1010

1010 XOR1100

=0110

Block 2

1010 XOR0110

=1100

1100 XOR1100

=0000

Block 3

1111 XOR0000

=1111

1111 XOR1100

=0011

Block 4

1010 XOR0011

=1001

1001 XOR1100

=0101

CBC Ciphertext

(b) Compare outputs

ECB CBC

0110 0110

0110 0000

0011 0011

0110 0101

Repeated plaintext produces repeated ciphertext in ECB.

CBC generates different ciphertext even for identical plaintext blocks.

(c) Pattern Analysis

ECB

- Same plaintext → Same ciphertext.
- Reveals patterns.
- Less secure.

CBC

- Each block depends on previous ciphertext.
- Repeated plaintext produces different ciphertext.
- Better confidentiality.

Conclusion: CBC hides patterns much better than ECB.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation